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Your Roll No. .....

CEN-503

B.Tech.(Computer Engineering) V Semester Examination 2024

Microprocessor

Paper No. : CEN-503

Time: 3 Hours

Maximum Marks: 45

(Write your Roll No. on top immediately on receipt of this question Paper)

Note: - Attempt all questions by attempting any two parts from each question.

CO1 Q1.

BL2 1(a). Draw and explain functional block diagram of 8085 microprocessor. 4.5

BL4 1(b). Distinguish between 8085 Microprocessor and 8086 Microprocessor. 4.5

BL2 1(c) Draw and explain timing diagram of IN 68 H instruction of 8085 stored at address 2300H. 4.5

CO2 **Q2.**

BL2 2(a). Explain the various flags available in 8086 microprocessor with their significance with suitable example. 4.5

BL3 2(b).

(i) Explain the function of the following program 1.5

```
      MVI  A , Byte
      ORA  A
      JM   NEXT
      OUT 68H
       HLT
NEXT: CMA
      ADI 01H
      OUT 68H
      HLT
```

(ii) What is the output at PORT1 when the following instructions are executed? Explain. 1.5

```
        MVI  A , 9FH
        ADI  82H
        JC   Display
        OUT  PORT1
        HLT
Display: XRA  A
        OUT  PORT1
        HLT
```

(iii) In the following program explain the range of the bytes that will be displayed at PORT2 1.5

```
        MVI  A , BYTE1
        MOV B, A
        SUI  60H
        JC   NEXT1
        MOV A , B
        SUI  90H
        JC   NEXT2
NEXT1 : XRA A
        OUT  PORT1
        HLT
NEXT2 : MOV  A , B
        OUT  PORT2
        HLT
```

BL3 2(c) Explain the meanings of the following instructions of 8085/8086 after writing their syntax. Also illustrate their use with the help of suitable example. 4.5

(i)SHLD

(ii)XCHG

(iii)LOOP

(iv)MOVSW

CO3 **Q3.**

BL6 3(a). Write a well documented program in the assembly language of 8085 to find HCF of given two integers 4.5

BL6 3(b) Write a program in assembly language of 8085 to arrange 400 integers stored at consecutive memory locations starting with address 2300H in Ascending order. Program should be well documented. 4.5

BL6 3 (c) Write a program in assembly language of 8085 to generate Modulo-09 counter with a delay of 60 ms between each count assuming a system with 0.5 microsecond clock period. Show all calculations. 4.5

CO4 Q4.

BL3 4(a). Explain how RIM and SIM instructions are used for interrupts in 8085 Microprocessor with the help of suitable example. 4.5

BL3 4(b).Explain how I/O devices are interfaced as Memory Mapped I/O with help of suitable diagram and examples. 4.5

BL3 4(c). What is PPI? How are peripheral are connected to microprocessor through 8255? Explain with the help of suitable example and diagrams. 4.5

CO5 Q5.

BL6 5(a). Write a program in assembly language of 8086 to find out the number of even number and no of odd numbers out of N 16 bit numbers stored at consecutive memory locations. 4.5

BL5 5(b) Can assembly language be used with C programming? If yes discuss how it can be done with suitable example. Also discuss how are software interrupt used in programming with suitable example. 4.5

BL2 5(c). Discuss the various addressing modes of 8086 Microprocessor with the help of suitable example. 4.5